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June 16, 1975
SUPERSEDING
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September 20, 1966

FEDERAL SPECIFICATION

COATING COMPOUND, CHEMICAL RESISTANT, CLEAR

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE

1.1 This specification covers one type of clear coating compound, which is resistant to concentrated acids.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

Federal Specifications:

TT-X-916 - Xylene (For Use in Organic Coatings).
PPP-B-585 - Boxes, Wood, Wirebound.
PPP-B-591 - Boxes, Shipping, Fiberboard, Wood-cleated.
PPP-B-601 - Boxes, Wood, Cleated-Plywood.
PPP-B-621 - Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636 - Box, Shipping, Fiberboard.
PPP-B-640 - Boxes, Fiberboard, Corrugated, Triple-Wall.
PPP-C-96 - Cans, Metal, 28 Gage and Lighter.
PPP-D-729 - Drums, Shipping and Storage, Steel, 55-Gallon.
PPP-P-704 - Pails, Metal: (Shipping, Steel, 1 through 12 Gallon).

Federal Standards:

Fed. Std. No. 123 - Marking for Shipment (Civil Agencies).
Fed. Test Method Std. No. 141/GEN - Paint, Varnish, Lacquer, and Related Materials; Methods of Inspection, Sampling, and Testing.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, DC, Atlanta, Chicago, Kansas City, MO, Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, WA.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

FSC 8010

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Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

American Society for Testing and Materials (ASTM) Standards:

D 1296 - Odor of Volatile Solvents and Diluents.
D 1475 - Density of Paint, Varnish, Lacquer, and Related Products.
D 1542 - Qualitative Tests for Rosin in Varnishes.
D 1544 - Color of Transparent Liquids (Gardner Color Scale).
D 1545 - Viscosity of Transparent Liquids by Bubble Time Method.
D 1644 - Nonvolatile Content of Varnishes.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

3. REQUIREMENTS

3.1 Material. The manufacturer is permitted wide latitude in the selection of raw materials and process of manufacture so that he may produce a coating compound for the intended uses.

3.2 Qualitative requirements.

3.2.1 Condition in container. When tested as specified in 4.4.3, the coating compound shall be free from skinning, suspended matter, sediment, and other undesirable qualities.

3.2.2 Skinning. When tested as specified in 4.4.3, the coating compound shall show no skinning.

3.2.3 Appearance. When tested as specified in 4.4.3, the coating compound shall be clear and free from haze, turbidity, and other suspended matter.

3.2.4 Reducibility with xylene. When tested as specified in 4.4.3, the coating compound shall mix readily and easily with xylene and shall show no curdling, precipitation, or separation.

3.2.5 Brushing properties. When tested as specified in 4.4.3, the coating compound shall brush easily, have good flowing and spreading qualities, and shall dry to a smooth, uniform film, free from seeds, runs, sags, or streaks. The dried film shall show no brush or lap marks.

3.2.6 Odor. When tested as specified in 4.4.3, the odor of the coating compound shall be characteristic of thinners for interior use and shall not be abnormally offensive or disagreeable.

3.2.7 Rosin and rosin derivatives. When tested as specified in 4.4.3, the coating compound shall contain no rosin or rosin derivatives.

3.2.8 Flexibility. When tested as specified in 4.4.3, the dried film of the coating compound shall show no evidence of cracking.

3.2.9 Knife test. When tested as specified in 4.4.3, the dried film of the coating compound shall adhere tightly to and shall not flake or crack from the metal. The film shall ribbon or curl from the metal on cutting and the cut shall show beveled edges.

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3.2.10 Hydrogen peroxide resistance. When tested as specified in 4.4.3, the dried film of the coating compound shall show no deterioration other than a slight whiting.

3.2.11 Acid resistance. When tested as specified in 4.4.3, the dried film of the coating compound shall show no deterioration other than a slight discoloration.

3.3 Quantitative requirements. When tested as specified in 4.4.3, the coating compound shall comply with the requirements of table I.

TABLE I. Quantitative requirements

Characteristics	Requirements	
	Minimum	Maximum
Nonvolatile matter (percent by weight)	20	---
Water (percent by weight)	---	0.5
Density (lb./gal.)	8.00	---
Viscosity (Gardner-Holdt)	Q	S
Color (Gardner)	---	2
Drying time (dry hard) (hours)	---	4

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Lot. For the purposes of sampling, a lot of the coating compound shall consist of a manufacturer's batch. The batch is defined as the end product of all raw materials mixed, blended, or processed in a single operation.

4.3 Inspection.

4.3.1 Inspection of filled containers. A random sample of filled containers shall be selected in accordance with MIL-STD-105 at inspection level I and acceptable quality level (AQL) of 2.5 percent defective. Each sample filled container shall be examined for defects of construction of the container, and the closure for evidence of leakage and for unsatisfactory markings; each filled container shall also be gauged to determine the amount to contents. Any container in the sample having one or more defects or under required fill shall be rejected, and if the number of defective containers in any sample exceeds the acceptance number for the appropriate sampling plan of MIL-STD-105, the lot represented by the sample shall be rejected.

4.3.2 Inspection of preparation for delivery requirements. An inspection shall be made to determine that the packaging, packing, and marking comply with the requirements of section 5. Defects shall be scored in accordance with table II. For examination of interior packaging, the sample unit shall be one shipping container fully prepared for delivery, selected at random just prior to the closing operations. Sampling shall be in accordance with MIL-STD-105. Defects of closure listed shall be examined on shipping containers fully prepared for delivery. The lot size shall be the number of shipping containers in the end item inspection lot. The inspection level shall be S-2 and AQL shall be 4.0 defects per hundred units.

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TABLE II. Classification of preparation for delivery defects

Examine	Defects
Markings (exterior and interior)	Omitted, improper size, location, sequence, or method of application.
Materials	Any component missing or damaged.
Workmanship	Inadequate application of components such as incomplete closure of container flaps, loose strapping, inadequate stapling. Bulging or distortion of container.

4.4 Testing of the end item.

4.4.1 Sampling for testing. For purposes of sampling, the lot shall be expressed in units of gallons of coating compound. The sample unit for testing shall be 1 quart of coating compound randomly selected from containers in the lot. The coating compound shall be placed in separate clean, dry metal or glass containers, sealed, marked, and forwarded to the testing laboratories. The sample size shall be as follows:

Lot size (gallons)	Sample size
800 or less	2
801 up to and including 22,000	3
22,001 and more	5

Tests shall be performed on a sample unit basis. All test reports shall contain the individual values utilized in expressing the final result. The lot shall be unacceptable if one or more sample unit fails to meet any test requirement specified.

4.4.2 Standard test conditions. Unless otherwise specified, all test specimens shall be prepared and tested in a room having a temperature of 23 ± 1° C. (73 ± 2° F.) and a relative humidity of 50 ± 5 percent.

4.4.3 Test methods. All tests shall be conducted in accordance with the methods specified in table III to determine compliance with the requirements of 3.2 and 3.3.

TABLE III. Test methods

Test	Methods		
	ASTM Standard	Fed. Test Method Std. No. 141	Paragraph reference
Condition in container	-	3011	-
Skinning	-	3021	4.4.4
Appearance	-	4261	-
Reducibility with xylene	-	4203	4.4.5
Brushing properties	-	4321	-
Odor	D 1296	-	-
Rosin and rosin derivitives	D 1542	-	-
Flexibility	-	6221	4.4.6
Knife test	-	6304	4.4.7
Hydrogen peroxide resistance	-	-	4.4.8
Acid resistance	-	-	4.4.9
Nonvolatile matter	D 1644	-	-
Water	-	4081	-
Density	D 1475	-	-
Viscosity	D 1545	-	-
Color	D 1544	-	-
Drying time	-	4061	-

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4.4.4 Skinning. Skinning shall be determined after 48 hours.

4.4.5 Reducibility with xylene. The xylene used shall conform to TT-X-916. The ratio of the volume of coating compound to the volume of xylene shall be one to one.

4.4.6 Flexibility. The coating compound shall be applied to a flat tin panel .009" x 3" x 8" with a 0.003-inch Bird film applicator (0.006-inch gap clearance) or similar doctor blade. The coating compound shall be dried for 2 hours in a horizontal position and then baked for 24 hours in an air-circulating oven at 100° to 103° C. At the end of the baking period the panel shall be allowed to stand for 30 minutes at standard conditions (see 4.4.2) and then bent at its midpoint through 180° over a 1/4-inch mandrel.

4.4.7 Knife test. The coating compound shall be applied to a flat tin panel .009" x 3" x 8" by brush so as to obtain a wet film thickness of 1 mil. The panel shall be air-dried for 18 hours, baked for 3 hours at 121° C. (250° F.), and cooled at standard conditions (see 4.4.2).

4.4.8 Hydrogen peroxide resistance. The coating compound shall be applied to a standard 3" x 9" cold rolled steel panel by dipping until a dry film thickness of 1 mil is obtained. Between dips, the panel shall be baked for one hour at 150° C. (302° F.) and cooled at standard conditions (see 4.4.2). Three drops of 30 percent hydrogen peroxide shall be placed on the dried film and covered with a watch glass with a diameter of 25 mm. After seven days, the watch glass and liquid shall be removed, and the panel shall be examined to determine compliance with the requirement of 3.2.10.

4.4.9 Acid resistance. The test panel shall be prepared as specified in 4.4.8. Three drops each of fuming sulfuric acid containing 30 percent by weight free SO_3 and white fuming nitric acid shall be placed in separate puddles on the dried film and covered with watch glasses with diameters of 25 mm. After 24 hours, the watch glasses and liquid shall be removed, and the panel shall be examined to determine compliance with the requirement of 3.2.11.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A, B, or C as specified (see 6.2).

5.1.1 Level A. The coating compound furnished in 1-gallon, 5-gallon or 55-gallon quantities as specified (see 6.2) shall be packaged in round cans conforming to PPP-C-96, type V, class 2; pails conforming to PPP-P-704, type II, class 3 or 4; or drums conforming to PPP-D-729, class I or II, as applicable. Exterior plan B shall be utilized for exterior coating of the cans in accordance with PPP-C-96.

5.1.2 Level B. The coating compound furnished in 1-gallon, 5-gallon, or 55-gallon quantities as specified (see 6.2) shall be packaged in accordance with 5.1.1, except plan A shall be utilized for exterior coating of the cans.

5.1.3 Level C. The coating compound shall be packaged in accordance with the supplier's commercial practice.

5.2 Packing. Packing shall be level A, B, or C as specified (see 6.2).

5.2.1 Level A. One-gallon cans of coating compound in the quantities as specified (see 6.2) shall be packed in close-fitting box conforming to PPP-B-585, class 3, PPP-B-591, class II, PPP-B-601, overseas type, PPP-B-621, class 2, PPP-B-636, class weather-resistant, or PPP-B-640, class 2. The boxes shall be closed and strapped in accordance with the applicable specification or appendix thereto. The gross weight of the triple wall boxes or the wood boxes shall not exceed 200 pounds. The gross weight of the PPP-B-636 boxes shall not exceed the weight limitations of the box specification. No packing shall be required for the 5-gallon pails or 55-gallon drums of the coating compound.

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5.2.2 Level B. Four one-gallon cans of the coating compound shall be packed in a close-fitting box conforming to PPP-B-636, class domestic. The box shall be closed in accordance with the appendix to PPP-B-636. No packing shall be required for the 5-gallon pails or 55-gallon drums of the coating compound.

5.2.3 Level C. The coating compound shall be packed to insure carrier acceptance and safe delivery at destination in containers complying with the rules and regulations applicable to the mode of transportation.

5.3 Marking.

5.3.1 Civil agencies. In addition to markings required by the contract or order, the cans and shipping containers shall be marked in accordance with Fed. Std. No. 123.

5.3.2 Military activities. In addition to markings required by the contract or order, the cans and shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. This coating compound is intended for use on interior or exterior metal in areas where maximum protection from concentrated acids is desired. The compound can be applied by spraying, dipping, or brushing. To spray, the coating compound should be reduced 50 percent with xylene. To dip or brush, reduce 25 percent with xylene. To achieve maximum resistance to strong acids it is recommended that 3 or 4 thin coats be applied rather than 1 or 2 heavy coats. This coating compound provides excellent adhesion over substrates as long as cleaning and dust free conditions are maintained.

6.2 Ordering data. Purchasers should select the preferred options permitted herein, and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Size and type of container required (see 5.1).
(c) Quantity required.
(d) Level of packaging and packing required (see 5.1 and 5.2).

6.3 The coating compound covered by this specification should be purchased by volume, the unit being one U.S. liquid gallon of 231 cubic inches at 15.5° C. (60° F.).

Preparing activity:

GSA-FSS

Orders for this publication are to be placed with General Services Administration, acting as an agent for the Superintendent of Documents. See Section 2 of this specification to obtain extra copies and other documents referenced herein. Price 30 cents each.

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